

Rajesh Sawant

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SUMMARY

Software Engineer with 3+ years across enterprise systems and applied AI. Currently building LLM-powered extraction pipelines processing 55,000+ manufacturer websites on SUDOKN, an NSF-funded knowledge graph platform at Arizona State University. Previously delivered client-facing enterprise integrations for manufacturing and automotive clients. Published at IEEE COMPSAC 2025.

EDUCATION

Arizona State University

Master of Science, Computer Software Engineering

• **GPA:** 3.76/4

• **Coursework:** Data Structures and Algorithms, Object Oriented Programming, Cloud Computing, Machine Learning, Big Data, Data Mining, Generative AI, Software Security, Testing and Validation, Software Project Management, Distributed Systems

Aug 2024 - May 2026

Tempe, AZ

Savitribai Phule Pune University

Bachelor of Engineering, Electronics and Telecommunications

Aug 2016 - Dec 2020

Pune, India

TECHNICAL SKILLS

- **Programming Languages:** Python, Java, SQL, TypeScript, JavaScript, C++, HTML, CSS
- **Frameworks :** React.js, FastAPI, Node.js, Next.js, JUnit, Spring Boot, Tailwind CSS, Jest, Playwright, Redux Toolkit, shadcn/ui
- **AI & Machine Learning :** Scikit-learn, TensorFlow, Pandas, NumPy, RAG, LangChain, LangGraph, Agentic AI, MCP, Prompt Engineering
- **Databases:** PostgreSQL, MySQL, MongoDB, Redis, GraphDB, Firestore, Neo4j, Supabase, Pinecone, FAISS
- **Cloud & DevOps:** Amazon Web Services (AWS), Google Cloud Platform (GCP), Microsoft Azure, Firebase, GitHub Actions, Docker
- **Tools & Technologies:** Jira, Confluence, Postman, Mendix, Figma, n8n, Cursor, Claude Code, Protégé, Ollama

WORK EXPERIENCE

Arizona State University

Graduate Service Assistant - Software Developer

Jun 2025 - Present

Tempe, AZ

- Built the manufacturer onboarding and validation system (Next.js/TypeScript, FastAPI, MongoDB) with role-based access and human-override workflows, providing reliable ground truth that improved the accuracy of the extraction pipeline
- Built responsive, accessible UI components with shadcn/ui and Tailwind CSS, improving usability across devices and increasing user satisfaction by reducing onboarding time and registration errors.
- Implemented LLM-powered real-time and batch extraction workflows with asynchronous workers, retries, and controlled concurrency, increasing structured data coverage while keeping extraction latency within production limits.
- Built a distributed, queue-driven ETL pipeline processing data from 55,000+ manufacturer websites, using AWS SQS to decouple ingestion, extraction, and reconciliation stages and improve system fault tolerance.
- Built an AI agent using Vercel's AI SDK that chains multiple tool calls per turn against the Proto-OKN MCP server to answer natural-language questions, and added a custom tool that renders answers as filtered map views, enabling users to obtain location insights quickly and improving the overall query response experience

Intelizign Lifecycle Services

Software Engineer

Aug 2021 - Jul 2024

Pune, India

- Designed and delivered enterprise web applications for manufacturing and automotive clients, collaborating with Scrum teams to turn business needs into scalable solutions that improved process efficiency and reduced onboarding time.
- Developed and managed data pipelines using REST APIs and service-oriented architecture to integrate and unify large datasets across Mendix, Java services, and external systems, increasing operational efficiency by 30% and supporting advanced analytics.
- Built reusable React UI components and contributed to UI/UX design improvements, including conducting UI/UX design workshops and onboarding sessions for new team members to improve development consistency and usability standards.
- Led Agile execution at the team level, contributing to sprint planning, backlog refinement, code reviews; containerized and deployed applications on AWS with Docker, contributing to CI/CD pipelines that reduced release turnaround.

PROJECTS

HazelAI - Agentic Hazard Intelligence

Jan 2026 - May 2026

- Led a 5-member team in delivering an AI-powered hazard intelligence platform, owning the agent workflow, backend integration, evaluation design, technical documentation, and coordination across frontend, graph, and data-pipeline workstreams.
- Designed a LangGraph-based AI agent that converts natural-language questions into executable SPARQL queries through context retrieval, query generation, execution feedback, and answer synthesis using LLM APIs.
- Built FastAPI services for query orchestration, validation, retry handling, and response generation, adding read-only execution controls and forced result limits to reduce risk from unsafe LLM-generated queries.

Sahana - Smart Event and Meetup Platform <https://sahana-drab.vercel.app/>

Jul 2024 - Present

- Engineered a full-stack event platform with FastAPI and React/TypeScript, implementing Google SSO authentication, JWT-based access control, cursor-based pagination across 12+ API endpoints, and a Redis caching layer with tiered TTLs that reduced redundant Firestore reads and improved query response times for paginated event feeds.
- Built an automated ETL ingestion pipeline triggered via GCP Cloud Scheduler, pulling from the Ticketmaster API across all user cities, deduplicating events through a two-tier strategy (Redis + Firestore fallback), and deploying to Cloud Run with GitHub Actions CI/CD for zero-downtime releases.
- Developed an interest and location-aware friend recommendation engine using weighted scoring (Jaccard similarity on interests, geodistance decay, and attended-category overlap), and built a performance-optimized React frontend with Redux Toolkit state management, Google Maps integration, infinite scroll, and smart cache invalidation on data mutations.